Namaa Community at IslamicEval 2026: Retrieval-Grounded Verification of Qur'anic and Hadith Citations for Hallucination Identification

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Abstract

We present the Namaa Community system for Subtask 2 of IslamicEval 2026, hallucination identification in Islamic citations generated by large language models. Given an Arabic response and its located citation segments, the system decides for each quoted verse (Ayah), hadith body (matn), chain of narration (isnad) and stated attribution (claimed source) whether it faithfully matches an authentic source. We treat the problem as retrieval-grounded verification: each segment is normalised, matched against the canonical Qur'an and the six hadith collections through a character n-gram index refined by edit-distance re-ranking, and adjudicated by a verifier chosen according to its type. A quoted Ayah or matn is verified by its similarity to the nearest authentic verse or narration; the attribution is then checked against the source its parent text matched, and the isnad is grounded in the parent hadith's complete narration. The system attains a macro accuracy of 0.846 on the development set and 0.668 on the official blind test. Our code, preprocessing pipeline and submissions are made available.[1]

1 Introduction

When large language models answer religious questions they frequently quote scripture—a verse of the Qur'an or a saying of the Prophet (a hadith)—and such quotations are a distinctive locus of hallucination: a model may alter a verse, misattribute a saying, or invent a chain of narrators. Unlike open-domain factuality, the ground truth here is finite and canonical, so faithfulness can be checked exactly. IslamicEval 2026 (Alharbi et al., 2026) formalises this over Arabic responses, continuing the inaugural edition (Mubarak et al., 2025).

We address Subtask 2, in which the citation spans are given and the system returns a verdict for each. A citation comprises two text segments, the Ayah and the hadith body (matn), and two structurally dependent segments, the chain of transmitters (isnad) and the stated attribution (claimed source); the isnad and claimed source are scored only when their parent text is correct. Systems are ranked by accuracy per type, macro-averaged over the four types (Alharbi et al., 2026), so a rare type weighs as much as a frequent one. We therefore verify all four types with equal care rather than optimising only the abundant text segments: Qur'anic verses are matched near word-for-word and hadith bodies with transmission tolerance, the attribution is checked against the source its parent text matched, and the isnad is grounded in the parent hadith. We report per-type results throughout, and find that the two structurally dependent types, being both rare and initially weak, are where balanced effort yields the largest returns.

2 Related Work

Verifying generated text against evidence is the concern of automated fact verification, from the FEVER benchmark (Thorne et al., 2018) to reference-free hallucination detectors that score factual precision or self-consistency (Manakul et al., 2023; Min et al., 2023); broader surveys place these within factuality evaluation for large language models (Ji et al., 2023), and retrieval augmentation is the standard mitigation (Lewis et al., 2020). Our setting differs in that the claims are exact quotations checked against fixed canonical sources, so verification reduces to grounded matching rather than open-ended

[1]https://github.com/NAMAA-ORG/Namaa-Community-at-IslamicEval-2026

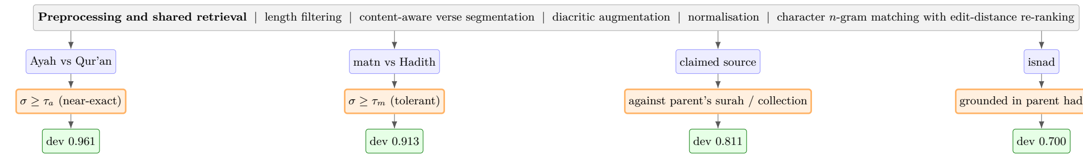

Figure 1: The Namaa Community pipeline. A shared preprocessing and retrieval stage grounds every segment in the canonical corpora; four typed verifiers produce the verdict. Green nodes report development accuracy per segment type (macro 0.846).

entailment. For Arabic, fact-checking has been approached through stance over retrieved evidence (Alhindi et al., 2021). The closest precedents are the inaugural IslamicEval systems (Mubarak et al., 2025): TCE (ElKoumy et al., 2025) and HUMAIN (Omayrah et al., 2025) report that Qur'anic quotations must match near word-for-word once diacritics are removed while hadith bodies require tolerance, an asymmetry we adopt; BurhanAI (Al Adel et al., 2025) verifies through a layered exact-to-semantic index whose cheaper tiers we reuse; and our earlier entry (Emad Eldin, 2025) targeted span detection rather than verification. Methodologically the pipeline builds on character n-gram term weighting (Salton and Buckley, 1988; Pedregosa et al., 2011), edit-distance re-ranking (Levenshtein, 1966; Bachmann, 2021), and corpora from Qur'anic question answering (Malhas and Elsayed, 2020).

3 Task and Data

Verification is grounded against the two corpora provided by the organisers: the canonical Qur'an and the six major hadith collections. The Qur'an comprises 6,236 verses and the hadith corpus 34,994 records, of which 31,811 carry a non-empty body; each hadith record also provides its complete narration, the chain and body together, which the isnad verifier uses for grounding (§4). The development split has 484 responses and 2,728 segments; after non-applicable rows are removed, the scored segments number 698 Ayah (222 correct, 476 incorrect), 588 matn (121, 467), 429 claimed source (283, 146) and 30 isnad (16, 14). The text types are predominantly incorrect, the scored attributions predominantly correct, and the isnad is thin yet carries a full quarter of the metric.

4 System Overview

A shared stage grounds each segment in the corpora, and a verifier chosen by segment type renders the verdict (Figure 1).

Preprocessing. So that an undiacritised quotation aligns with a vocalised source, we prepare both corpora identically. Records of extreme length are discarded; any text over twenty-five sub-word tokens (Antoun et al., 2020) is split into at most two parts at the whitespace nearest its midpoint, so a partially quoted verse can match without breaking a word; every text keeps its vocalised original and gains a diacritic-free copy, roughly doubling the effective corpus; and each text is expanded into overlapping word windows to recover fragmentary quotations. A single normaliser is applied to corpus and query alike, removing diacritics and the elongation mark, unifying alef, ya, waw-hamza and ta-marbuta variants, and collapsing non-Arabic characters, with its diacritic ranges specified by Unicode code point (Appendix E). Examples appear in Appendix D.

Retrieval. For each corpus we build a character n-gram index over three- to five-character grams (Pedregosa et al., 2011), whose sub-word units resist Arabic inflection; a query returns a shortlist that is re-ranked by the higher of an order-insensitive and a substring-alignment edit-distance measure (Bachmann, 2021). We write $\sigma(x)$ for the similarity of a span x to its best candidate.

Text segments. An Ayah or matn is judged by thresholding σ: correct when $\sigma(x) \geq \tau_t$ and incorrect otherwise, with $t \in \{a, m\}$. The thresholds differ, following the asymmetry noted above: a Qur'anic quotation must be near-exact, so τ_a is high, whereas a matn admits transmission variation, so τ_m is lower.

Claimed source. An attribution—a surah name and verse number, or a collection—is a reference, not quoted scripture, so matching it against the corpus is ill-posed. We retain, for each annotation, the record its parent Ayah or matn matched, and verify the attribution against that record: whether the surah and verse, or the collection, named agrees with the parent's source. The verifier is anchored to the majority label, returning correct unless a mismatch is detected.

Isnad. Because an isnad is scored only with a correct matn, the parent matn has matched a hadith record whose complete narration contains the authentic chain. We ground the quoted isnad by its similarity to that narration, over the parent's strongest matches, thresholded at τ_i, replacing the majority prior a verifier without the matched source would need. A verdict is emitted for every segment; non-applicable segments are excluded by the scorer, so none is left unpredicted.

5 Experiments

5.1 Setup

The thresholds τ_a, τ_m and τ_i are the only fitted quantities. They are selected on a sample of 1,200 responses from the training split by maximising macro accuracy, then frozen ($\tau_a = 0.98$, $\tau_m = 0.94$, $\tau_i = 0.85$) and applied unchanged to the evaluation splits; fitting on training rather than on the evaluation data keeps the reported figures an honest estimate of generalisation. Every figure is produced by the organisers' official scoring.

5.2 Results

Table 1 reports the development ablation, each row adding one component, together with the official blind-test submission. From a configuration that matches the attribution against the corpus and defaults the isnad to its majority label, linking the attribution to its parent record raises that type from 0.492 to 0.811 and the macro average by eight points; grounding the isnad then raises it from 0.533 to 0.700 and the macro average by a further four, to 0.846. The two frequent text types are already strong (Ayah 0.961, matn 0.913) and are unchanged by these steps, so the improvement is carried almost entirely by the two structural types, as the macro metric predicts. Appendix F compares the character n-gram retriever against word-level TF-IDF and BM25 backends, and Appendix G gives representative per-type misclassifications.

5.3 Qualitative Analysis

Three cases illustrate the verifiers. A verse quoted verbatim reaches a similarity close to unity and is labelled correct, whereas one in which a single word has been substituted falls below τ_a and is labelled incorrect, whereupon its attribution becomes non-applicable and is excluded. A hadith body correctly quoted but attributed to البخاري while its matched record belongs to مسلم is caught by the parent-linked verifier, which returns incorrect on the collection mismatch even though the body itself is authentic. A correct body accompanied by a chain whose similarity to the parent hadith's narration reaches 0.90 clears τ_i and is labelled correct; this is the mechanism behind the strong isnad accuracy on the blind test.

5.4 Error Analysis

The blind-test macro of 0.668 is lower than on development, but its per-type profile is informative rather than uniformly depressed. The isnad rises to 0.895—grounding generalises, and the blind test has a larger, more separable isnad population—and the Ayah remains strong at 0.818; the decline concentrates in the matn (0.622) and the claimed source (0.340). The claimed-source figure falls far below its development value of 0.811, indicating that the parent-linked verifier did not transfer to the blind test—consistent with the scored submission not reflecting the completed configuration and with a shifted attribution distribution. The matn decline is consistent with hadith quotations drawn more widely across the six collections than the development sample, for which the overlapping-window expansion is the intended countermeasure. Errors are also coupled: since the attribution and isnad verifiers depend on the record matched by the parent Ayah or matn, a retrieval miss on the parent propagates to its dependents.

6 Discussion

Two observations generalise beyond this task. First, when an evaluation macro-averages over

System configuration	Ayah	matn	claimed source	isnad	Macro
Development ablation (each row adds one component)					
Attribution matched as text (initial)	0.961	0.913	0.492	0.533	0.725
+ parent-linked attribution	0.961	0.913	0.811	0.533	0.805
+ grounded isnad (submitted)	0.961	0.913	0.811	0.700	0.846
Official blind test					
Submitted system	0.818	0.622	0.340	0.895	0.668

Table 1: Per-segment-type accuracy and macro average on the development set (upper panel, each row cumulatively adding one component to the previous) and on the official blind test (lower panel). Missing predictions were zero throughout.

segment types of very different frequency, the rare types govern the attainable score; our gains came from the isnad and attribution verifiers rather than the abundant text types, and aggregate factuality scores can likewise mask weakness on infrequent claim types (Min et al., 2023). Second, exact-quotation verification against a closed canon is a distinct and tractable regime: unlike open-domain fact verification (Thorne et al., 2018) or reference-free hallucination detection (Manakul et al., 2023), the evidence is fixed and complete, so a transparent grounding pipeline suffices and remains auditable—a desirable property for religious content, where an opaque judgement is hard to defend. The gap between our development and blind-test scores adds a practical corollary: the configuration validated offline must be the one submitted, and per-type diagnostics are what localise degradation under distribution shift.

7 Conclusion

The Namaa Community system verifies Islamic citations by grounding each in the canonical corpora, and reads the macro-averaged metric as an instruction to invest in the rare structural segment types. An attribution check against the citation's parent source and an isnad verifier grounded in the parent hadith raise development macro accuracy from 0.725 to 0.846, with essentially all of the gain in those two types. The official blind-test result of 0.668 and its per-type decomposition localise the remaining work to the matn and attribution verifiers, and motivate resubmission of the completed configuration.

Limitations

The development analysis rests on a single split, and its isnad figure is estimated from only thirty scored instances; the blind test, with far more isnad segments, is the more reliable estimate for that type. The verifiers are coupled through retrieval, so an Ayah or matn that fails to match will also mislead the dependent attribution and isnad checks. Isnad grounding relies on the complete-narration content of the hadith records and would be strengthened by an explicit narrator database. Thresholds are transferred without per-split adaptation. Finally, a system validated on development is not automatically the one reflected in a scored submission; the reported blind-test figure is the official one, and closing the gap it exposes is left to the next cycle.

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A Per-type Development Scores

On the development set the submitted configuration attains a macro accuracy of 0.846, with per-type accuracies of 0.961 for the Ayah, 0.913 for the matn, 0.811 for the claimed source and 0.700 for the isnad, and no missing predictions. The corresponding blind-test values appear in the lower panel of Table 1.

B Isnad Verifier

Grounding the quoted chain in the parent hadith's complete narration exceeds the majority prior by nearly seventeen points on development, and transfers more stably from training than grounding in the chain alone (Table 2). On development the similarity separates the classes, averaging 0.84 for correct chains against 0.75 for incorrect ones.

Isnad verifier	Train acc.	Dev acc.
Majority prior	—	0.533
Grounded in chain only	0.663	0.700
Grounded in full narration	0.719	0.700

Table 2: Isnad verification strategies.

C Thresholds and Retrieval Settings

The retriever indexes three- to five-character grams and returns a fifteen-candidate shortlist; the re-ranking similarity is the maximum of an order-insensitive and a substring-alignment edit-distance score, normalised to $[0, 1]$. The thresholds, fitted on a 1,200-response training sample, are $\tau_a = 0.98$, $\tau_m = 0.94$ and $\tau_i = 0.85$, and isnad grounding considers the three strongest parent-matn matches.

D Corpus Preprocessing Examples

Table 3 illustrates the transformations of §4. The originals are always retained; every transformation adds indexable variants rather than replacing the source.

Transformation	Illustration
Segmentation of over-length verses	a long verse is divided into two parts at the whitespace nearest its midpoint, with no word broken
Diacritic augmentation	the vocalised original is kept and an undiacritised copy added, e.g. الحمد لله رب العالمين alongside its fully marked form
Overlapping windows	a twenty-word body yields windows of five to fifteen words over both the original and normalised forms

Table 3: Preprocessing transformations with illustrations.

E Normalisation of Arabic Diacritic Ranges

The normaliser's diacritic-removal ranges are specified numerically, by Unicode code point, rather than by writing the Arabic combining marks literally. Literal combining marks do not render as standalone glyphs and can reorder relative to the delimiter of a character range when a source file is saved, silently widening the intended range so that it comes to include the base Arabic letters; the normaliser would then delete all Arabic text and every span would fail to match. Specifying the ranges numerically removes this failure mode, which is otherwise invisible on inspection yet fatal to the result.

F Retrieval Backend Comparison

To isolate the effect of the candidate retriever from the shared re-ranking and verifiers, we swap the character n-gram index for a word-level TF-IDF index and for Okapi BM25, keeping every other component and the per-backend tuned thresholds fixed, and re-score the development set with the official metric. Table 4 reports the result.

G Misclassified Development Examples

Table 5 shows one representative misclassification per segment type on the development set, with the quoted span, the gold and predicted labels, and the nearest canonical source retrieved.

Retrieval backend	Ayah	matn	c. src	isnad	Macro
character n-gram TF-IDF (ours)	0.961	0.913	0.811	0.700	**0.846**
word-level TF-IDF	0.961	0.912	0.823	0.667	0.841
Okapi BM25	0.963	0.927	0.823	0.667	0.845

Table 4: Development macro accuracy with only the candidate retriever swapped, every other component and the per-backend tuned thresholds held fixed. Character n-gram TF-IDF attains the best macro; BM25 is marginally behind, with a stronger matn but a weaker isnad, and word-level TF-IDF trails on isnad. “c. src” is the claimed source.

Type	gold/pred	quoted span	nearest source
Ayah	incorrect/correct	وَقَالَ رَبُّكُمُ ادْعُونِي أَسْتَجِبْ لَكُمْ	وَقَالَ رَبُّكُمُ ادْعُونِي أَسْتَجِبْ لَكُمْ ۚ إِنَّ ا...
matn	correct/incorrect	إن الله يرضى لكم ثلاثًا: أن تعبدوه ولا تشركوا به شيئًا،...	إِنَّ اللَّهَ يَرْضَى لَكُمْ ثَلَاثًا ، وَيَكْرَهُ لَكُمْ...
isnad	correct/incorrect	عن علي رضي الله عنه قال:	كُنْتُ رَجُلًا مَذَّاءً ، وَكُنْتُ أَسْتَحْيِي أَنْ أَس...
claimed src	incorrect/correct	السورة 3، آية 139	فَإِذَا بَلَغْنَ أَجَلَهُنَّ فَأَمْسِكُوهُنَّ بِمَعْرُو...

Table 5: Representative development misclassifications, one per segment type.